CLASS 10 SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

Chapter 1 • Chemistry Lecture Notes

Study focus

Chemical reactions • Chemical equations • Balancing equations • Types of reactions • Oxidation and reduction • Corrosion • Rancidity • Exam practice

CBSE / NCERT-oriented revision notes

1. What is a chemical reaction?

A **chemical reaction** is a process in which one or more substances change into new substances with different properties.

• **Reactants:** Substances that take part in a reaction.

• **Products:** New substances formed during the reaction.

Example: Magnesium burns in oxygen to form magnesium oxide.

$$2Mg + O_2 \rightarrow 2MgO$$

Signs of a chemical reaction

• Change in colour — iron placed in copper sulphate solution changes the blue solution towards green.

• Evolution of gas — zinc reacts with dilute hydrochloric acid and releases hydrogen gas.

$$Zn + 2HCl \rightarrow ZnCl_2 + H_2$$

• Formation of a precipitate — sodium sulphate and barium chloride solutions form insoluble barium sulphate.

$$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow + 2NaCl$$

• Change in temperature or state may also indicate a reaction.

2. Chemical equations

A chemical equation represents a chemical reaction using symbols and chemical formulae.

Word equation: Magnesium + Oxygen → Magnesium oxide

Unbalanced equation:

$$Mg + O_2 \rightarrow MgO$$

Balanced equation:

$$2Mg + O_2 \rightarrow 2MgO$$

Why balance chemical equations?

The **law of conservation of mass** states that mass can neither be created nor destroyed in a chemical reaction. Therefore, each element must have the same number of atoms on both sides of a balanced equation.

Steps to balance an equation

Example: Hydrogen reacts with oxygen to form water.

$$H_2 + O_2 \rightarrow H_2O$$

$$2H_2 + O_2 \rightarrow 2H_2O$$

Balance atoms by changing the coefficients in front of formulae. **Never change subscripts** within a chemical formula, because doing so changes the substance.

3. Types of chemical reactions

A. Combination reaction

Two or more substances combine to form a single product. General form: A + B → AB.

$$2H_2 + O_2 \rightarrow 2H_2O$$

$$C + O_2 \rightarrow CO_2$$

$$CaO + H_2O \rightarrow Ca(OH)_2 + \text{heat}$$

The last reaction releases heat and is also an **exothermic reaction**.

B. Decomposition reaction

One compound breaks down into two or more simpler substances. Decomposition may be caused by heat, electricity or sunlight.

• **Thermal decomposition:**

$$CaCO_3 \xrightarrow{\text{heat}} CaO + CO_2$$

• **Electrolytic decomposition:**

$$2H_2O \xrightarrow{\text{electricity}} 2H_2 + O_2$$

• **Photochemical decomposition:**

$$2AgCl \xrightarrow{\text{sunlight}} 2Ag + Cl_2$$

Silver chloride changes from white to grey because silver is formed.

C. Displacement reaction

A more reactive element displaces a less reactive element from its compound.

$$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$$

Iron displaces copper from copper sulphate solution. The blue solution becomes greenish and a reddish-brown copper deposit forms on the iron.

D. Double displacement reaction

Two compounds exchange ions to form two new compounds.

$$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow + 2NaCl$$

The white solid barium sulphate is a **precipitate**. A reaction that forms an insoluble solid is called a precipitation reaction.

E. Redox reaction

Oxidation and reduction occur simultaneously in a redox reaction.

$$CuO + H_2 \rightarrow Cu + H_2O$$

• **Oxidation:** Addition of oxygen or removal of hydrogen.

• **Reduction:** Removal of oxygen or addition of hydrogen.

In this reaction, copper oxide loses oxygen and is reduced to copper; hydrogen gains oxygen and is oxidised to water.

4. Exothermic and endothermic reactions

Exothermic reaction

A reaction that releases energy, usually as heat, is called exothermic.

$$CaO + H_2O \rightarrow Ca(OH)_2 + \text{heat}$$

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{heat}$$

Endothermic reaction

A reaction that absorbs energy from its surroundings is called endothermic. Many decomposition reactions need heat, light or electricity to proceed.

5. Oxidation in everyday life

Corrosion

Corrosion is the gradual damage of a metal due to reactions with substances in its environment. Rusting is the corrosion of iron in the presence of oxygen and moisture. Rust is hydrated iron(III) oxide.

Prevention: Painting, oiling or greasing, galvanising, electroplating and using suitable alloys.

Rancidity

Rancidity occurs when fats and oils in food are oxidised, producing unpleasant smell and taste.

Prevention: Store food in airtight containers, refrigerate it, keep it away from light, use antioxidants where appropriate, or pack it under nitrogen in suitable products.

6. Important equations to revise

2Mg + O■ → 2MgO

2H■ + O■ → 2H■O

CaO + H■O → Ca(OH)■ + heat

CaCO■ —heat→ CaO + CO■

2AgCl —sunlight→ 2Ag + Cl■

Fe + CuSO■ → FeSO■ + Cu

Na■SO■ + BaCl■ → BaSO■↓ + 2NaCl

Zn + 2HCl → ZnCl■ + H■

CuO + H■ → Cu + H■O

CH■ + 2O■ → CO■ + 2H■O

7. Exam practice

Very short-answer questions

1. What are reactants and products?

2. State the law of conservation of mass.

3. What is a precipitate?

4. Define corrosion and rancidity.

Short-answer questions

5. Balance: H■ + O■ → H■O.

6. Explain the difference between combination and decomposition reactions, with one equation each.

7. What observations occur when an iron nail is placed in copper sulphate solution?

8. Explain oxidation and reduction using CuO + H■ → Cu + H■O.

Answer key

1. Reactants are starting substances; products are substances formed.

2. Mass is neither created nor destroyed in a chemical reaction.

3. An insoluble solid formed during a reaction in solution.

4. Corrosion is the gradual damage of metals by their environment; rancidity is oxidation of fats and oils in food.

5. 2H■ + O■ → 2H■O.

6. Combination: 2H■ + O■ → 2H■O. Decomposition: CaCO■ —heat→ CaO + CO■.

7. The blue solution becomes greenish and copper deposits on the nail.

8. CuO loses oxygen and is reduced; H■ gains oxygen and is oxidised.

Study tip: Practise balancing equations by counting atoms on each side, and learn the observations and conditions for the standard reactions.